



- Define computer memory and distinguish between primary and secondary memory.
- Explain the function of primary memory, including its types (RAM and ROM) and their characteristics.
- Differentiate between types of secondary memory (HDD, SSD, external storage, and cloud storage) and their key features.



UNIT 2

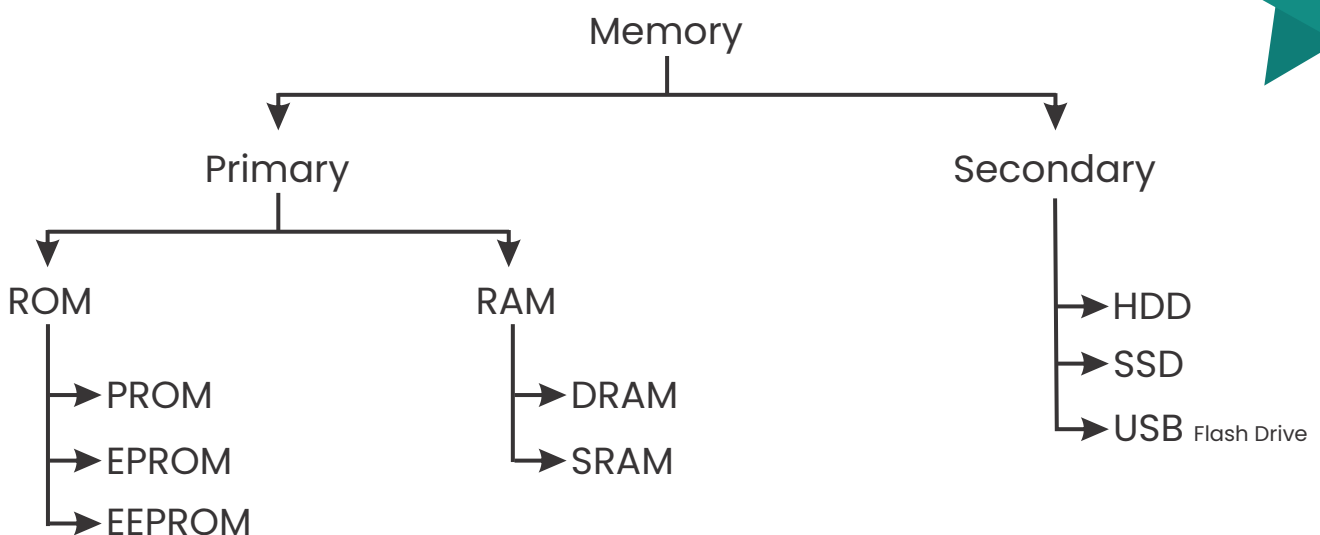
COMPUTER MEMORY

In this chapter, we will explore the two essential types of computer memory: Primary Memory and Secondary Memory. Understanding these memory types is crucial for grasping how computers store and manage data. Let's learn more about each type and their importance in the world of computing!



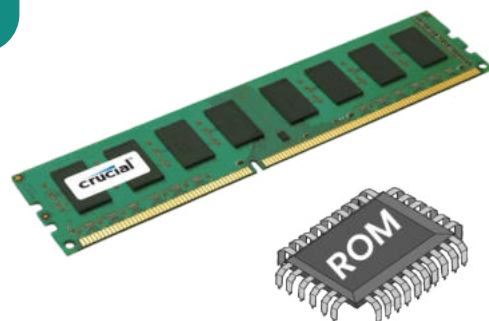
WHAT IS MEMORY?

Memory is a component of a computer that stores data and instructions. It allows the computer to quickly access and process information. There are two main categories of memory: **Primary Memory** and **Secondary Memory**.



PRIMARY MEMORY

Primary Memory, also known as volatile memory, is the memory that a computer uses to store data temporarily while it is in use.





UNIT 2

COMPUTER MEMORY

KEY FEATURES OF PRIMARY MEMORY:

1. **Volatility:** Data stored in primary memory is lost when the computer is turned off.
2. **Speed:** Primary memory is much faster than secondary memory, allowing quick access to data for processing.
3. **Types of Primary Memory:**
 - **Random Access Memory (RAM):** This is the main type of primary memory used to store data that is actively being used by the CPU.
 - **Read-Only Memory (ROM):** This type of memory contains important instructions and data that do not change.

IMPORTANCE OF PRIMARY MEMORY:

- **Performance:** A larger amount of RAM allows for smoother multitasking and faster processing speeds.
- **Temporary Storage:** Primary memory holds data that the CPU needs immediately, making it crucial for the computer's operation.



SECONDARY MEMORY

Secondary Memory, also known as non-volatile memory, is used to store data permanently, even when the computer is turned off.

KEY FEATURES OF SECONDARY MEMORY:

1. **Non-volatility:** Data stored in secondary memory remains intact even after the computer is powered down.
2. **Capacity:** Secondary memory generally has a much larger storage capacity compared to primary memory.



UNIT 2

COMPUTER MEMORY

TYPES OF SECONDARY MEMORY:

- **Hard Disk Drive (HDD):** A traditional storage device that uses spinning disks to read and write data. HDDs have a large capacity but are slower compared to SSDs.
- **Solid State Drive (SSD):** A more modern type of storage that uses flash memory to store data. SSDs are faster than HDDs and are becoming increasingly popular.
- **External Storage Devices:** USB drives and external hard drives are portable storage options that allow users to save and transfer data easily.
- **Cloud Storage:** This is an online storage solution where data is saved on remote servers. It allows you to access files from anywhere with an internet connection (e.g., Google Drive, Dropbox)



IMPORTANCE OF SECONDARY MEMORY:

- **Data Storage:** Secondary memory is where you keep your important files, photos, games, and documents. It is essential for saving your work.
- **Backup and Recovery:** Secondary memory allows for backup copies of data, ensuring you don't lose important information.



UNIT 2

COMPUTER MEMORY

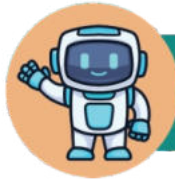


COMPARISON OF PRIMARY AND SECONDARY MEMORY

Feature	Primary Memory	Secondary Memory
Volatility	Volatile (data lost when off)	Non-volatile (data retained)
Speed	Faster	Slower
Capacity	Smaller	Larger
Types	RAM, ROM	HDD, SSD, External Drives, Cloud
Usage	Temporary data storage	Permanent data storage

Both primary and secondary memory play crucial roles in how computers function. Primary memory allows for quick access to active data, while secondary memory provides long-term storage for files and information.

By understanding these two types of memory, you will be better equipped to use and manage your computer effectively.



EXERCISE 1



UNIT 2

COMPUTER MEMORY

Choose the correct answer.

1. What is the primary function of RAM in a computer?

A Permanent storage of data

B Temporary storage of data

C Backup storage

2. Which of the following is a type of secondary memory?

A RAM

B ROM

C Hard Disk Drive (HDD)

3. What does ROM stand for?

A Read-Only Memory

B Random Access Memory

C Real-Time Operating Memory

4. Which memory type is faster?

A Secondary Memory

B Primary Memory

C Both are equally fast

5. What happens to the data in RAM when the computer is turned off?

A It is saved permanently

B It is lost

C It is transferred to secondary memory



UNIT 2

COMPUTER MEMORY

6. Which of the following is an example of cloud storage?

A USB flash drive

B Google Drive

C External Hard Drive

7. What is the main advantage of SSD over HDD?

A Larger storage capacity

B Lower cost

C Faster data access speeds

8. Which memory is used to store firmware?

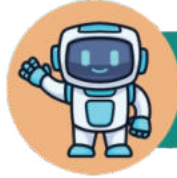
A RAM

B ROM

C HDD

DO YOU KNOW?

Firmware is a special kind of software inside devices like printers, cameras, or computers. It helps the device work properly by giving it basic instructions to follow. It is typically stored in non-volatile memory such as ROM, EPROM, or flash memory, and it is not meant to be modified frequently.



EXERCISE 2



UNIT 2

COMPUTER MEMORY

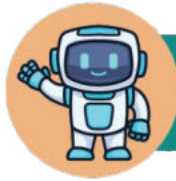
Fill in the blanks.

1. RAM is considered _____ memory because it loses data when the computer is turned off.
2. _____ drives are used for permanent data storage and are slower than RAM.
3. The _____ memory is often used to store the computer's operating system and applications.
4. When a computer is powered down, any data in RAM is _____.
5. The main advantage of using SSDs is their _____ access speed compared to HDDs.
6. _____ memory is essential for running applications and processing data quickly.



UNIT 2

COMPUTER MEMORY



EXERCISE 3

Mark "True" or "False".

TRUE

FALSE

RAM is a type of secondary memory.

☐☐

Data in primary memory is lost when the computer is turned off.

☐☐

HDDs are faster than SSDs.

☐☐

Cloud storage allows you to access files from any device with an internet connection.

☐☐

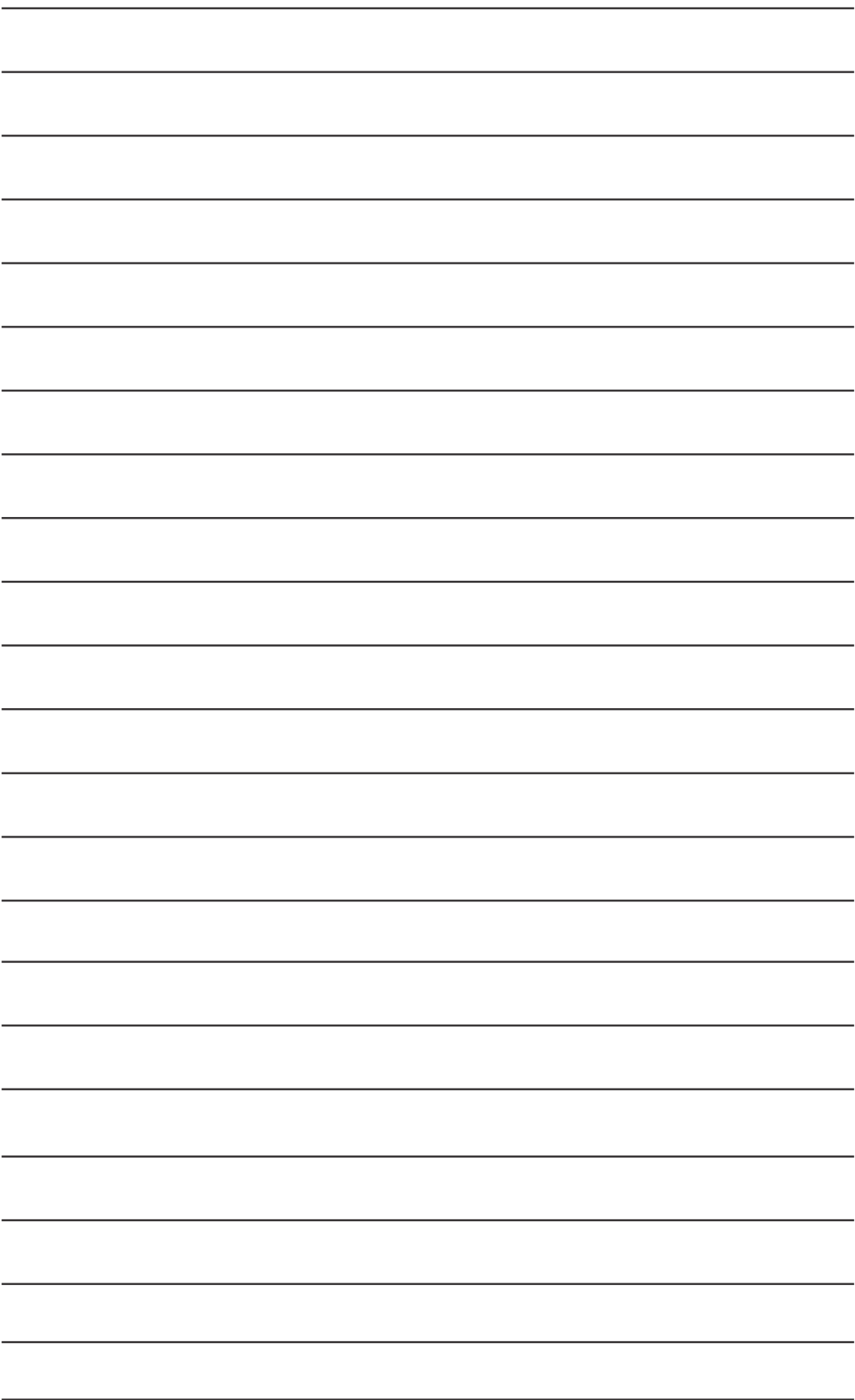
ROM is used for the temporary storage of data.

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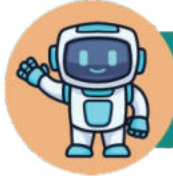


6. What is Cloud Storage?

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COMPUTER MEMORY



EXERCISE 5



UNIT 2

COMPUTER MEMORY

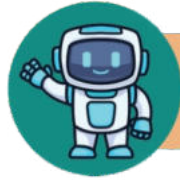
Match the following terms with their descriptions.

Term	Description
RAM	Primary memory for active data
ROM	Faster than HDD, uses flash memory
HDD	Stores the computer's firmware
SSD	Access files online from any internet-connected device
Cloud Storage	Permanent storage of data



UNIT 2

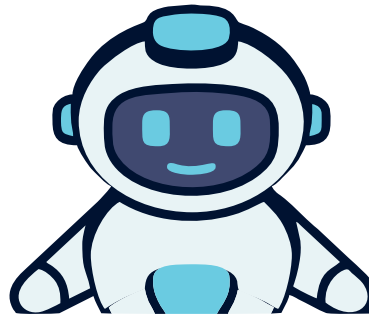
COMPUTER MEMORY



LAB ACTIVITY



THE LIBRARY OF DATA



Demonstrate how data is transferred between secondary and primary memory in a computer.

1. Setup: Create a “Library” (secondary memory) with books/files for long-term storage and a “Desk” (primary memory) for active work.

2. Role Play: Assign roles—Library Keeper (fetches books from the library) and Worker (processes tasks on the desk with limited space, like RAM).

3. Task Rotation: The Worker completes tasks (writing, sorting) using books from the desk, and then the Library Keeper returns them.